Exponentials & Logarithms

ANSWER KEY

Section 1: Exponential Equations

1 Solve for x: $3^{2x-1} = 27$.

a. $x = 2$ ✓

b. $x = 3$

c. $x = 1$

d. $x = 4$

$27 = 3^3$. Since bases are equal: $2x - 1 = 3 \Rightarrow 2x = 4 \Rightarrow x = 2$.

2 If $4^{x+1} = \left(\frac{1}{2}\right)^{x-4}$, what is x?

a. $x = 2$ ✓

b. $x = 1$

c. $x = 3$

d. $x = -2$

Express with base 2: $(2^2)^{x+1} = (2^{-1})^{x-4} \Rightarrow 2^{2x+2} = 2^{-x+4} \Rightarrow 2x + 2 = -x + 4 \Rightarrow 3x = 2 \Rightarrow x = 2$.

3 What is the horizontal asymptote of the exponential function $f(x) = 2^x - 5$?

a. $y = -5$ ✓

b. $y = 5$

c. $y = 0$

d. $x = -5$

As $x \to -\infty$, $2^x \to 0$, so $f(x) \to -5$. The horizontal asymptote is $y = -5$.

Section 2: Logarithmic Properties

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4 What is the value of $\log_2(32)$?

a. 5 ✓

b. 4

c. 6

d. 16

Since $2^5 = 32$, $\log_2(32) = 5$.

5 Evaluate $\log_3(54) - \log_3(2)$.

a. 3 ✓

b. 2

c. 4

d. 27

Quotient property: $\log_3(54) - \log_3(2) = \log_3\left(\frac{54}{2}\right) = \log_3(27) = 3$.

6 Simplify: $2\log_5(x) + \log_5(y)$.

a. $\log_5(x^2y)$ ✓

b. $\log_5(2xy)$

c. $\log_5(x^2 + y)$

d. $\log_5(xy^2)$

Power and product rules: $2\log_5(x) + \log_5(y) = \log_5(x^2) + \log_5(y) = \log_5(x^2y)$.

Section 3: Logarithmic Equations

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7 Solve for x: $\log_4(2x+8)=3$.

a. $x=28$ ✓

b. $x=24$

c. $x=32$

d. $x=16$

Convert to exponential form: $2x+8=4^3=64 \Rightarrow 2x=56 \Rightarrow x=28$.

8 Solve for x: $\log_2(x)+\log_2(x-2)=3$.

a. $x=4$ ✓

b. $x=2$

c. $x=-2$

d. $x=6$

$\log_2[x(x-2)]=3 \Rightarrow x^2-2x=2^3=8 \Rightarrow x^2-2x-8=0 \Rightarrow (x-4)(x+2)=0$. Since $\log$ requires $x>2$, $x=4$.

9 What is the domain of the function $f(x)=\log_3(4-x)$?

a. $(-\infty, 4)$ ✓

b. $[4, \infty)$

c. $(4, \infty)$

d. $(-\infty, 4]$

The argument of a logarithm must be strictly positive: $4-x>0 \Rightarrow x<4$, which is $(-\infty, 4)$.

Section 4: Natural Logarithms & Base e

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10 Evaluate $\ln(e^7)$.

a. 7 ✓

b. e

c. 1

d. 14

By the inverse property of natural logs: $\ln(e^x) = x$, so $\ln(e^7) = 7$.

11 Solve the natural exponential equation: $e^{2x} = 5$.

a. $x = \frac{\ln 5}{2}$ ✓

b. $x = 2\ln 5$

c. $x = \ln(2.5)$

d. $x = \frac{5}{2e}$

Take $\ln$ of both sides: $2x = \ln 5 \Rightarrow x = \frac{\ln 5}{2}$.

12 What is the value of $e^{\ln(12)}$?

a. 12 ✓

b. 1

c. e

d. 144

By definition of inverse operations: $e^{\ln x} = x$, so $e^{\ln(12)} = 12$.

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